

Midterm 2

ORF 309, Fall 2025

Instructions: This midterm has 3 questions for a total of 90 points. Partial credit will be awarded for partial proofs or arguments. Strive to reduce answers as much as possible (e.g., give a fraction in simplest terms, though a numerical answer requiring a calculator is not necessary). You may freely use basic facts from class, such as laws of probability, properties of conditional expectation, or manipulations with pmfs/pdfs. However, you must provide adequate justifications for important steps. Put another way, try your best to demonstrate awareness that you are applying the correct facts in the correct way by explicitly mentioning the results you are applying, indicating the objects they are applied to, and checking the required assumptions.

Length: 105 minutes (for working on the problems) + 15 minutes (for submission)

Good Luck!

Question	Points	Score
1	30	
2	30	
3	30	
Total:	90	

1. (Insurance) When customers file claims, the insurer's payment depends not only on the loss size but also on a variety of unpredictable factors such as regional regulations, claim-processing backlog, and negotiated discounts with repair shops and medical centers, etc. As a result, the fraction of each loss that the policyholder (the insured) must pay out-of-pocket varies randomly from claim to claim. Assume:

- X = total loss amount (in thousands of dollars) with tail distribution

$$\mathbb{P}(X > x) = \begin{cases} 1, & x < 1, \\ \frac{1}{x^2}, & x \geq 1, \end{cases}$$

- U = out-of-pocket fraction, uniformly distributed in $[0, 1]$, independent of X .

The insured individual pays $Y = UX$ (in thousands of dollars). For example, if the insured individual gets a medical bill with total cost $X = 10$ thousand dollars, and the out-of-pocket fraction turns out to be $U = 0.1$, then the individual pays $Y = UX = 0.1 \times 10 = 1$ thousand dollars.

- (a) (10 points) On average, how much does the insured individual expect to pay?
- (b) (10 points) Determine the joint probability density function of (X, Y) . (Hint: you may want to start with the conditional density of Y given X . Be careful of the range.)
- (c) (10 points) Suppose that after an incident the insured individual pays exactly 1 thousand dollars out-of-pocket. What is the probability that the total loss **exceeds** 2 thousand dollars?

2. (Sleepy Gambler) A gambler is playing a game on a one-dimensional board with positions 0, 1, 2, 3, 4, where

- position 0 means bankruptcy (the gambler loses everything), and
- position 4 means success (the gambler leaves the casino rich).

At each round, if the gambler is currently at position i (with $1 \leq i \leq 3$):

- she moves up to $i + 1$ with probability $\frac{1}{2}(1 - \alpha_i)$,
- she moves down to $i - 1$ with probability $\frac{1}{2}(1 - \alpha_i)$, and
- she stays idle (takes a nap) with probability α_i .

Here we take

$$\alpha_i = \frac{i}{4}$$

to be the “laziness parameter”, which depends on her current state i . Intuitively, the closer she is to success, the more time she needs to prepare herself for the next play. Once she reaches 0 or 4, the game ends.

- (10 points) Suppose the gambler starts at position 2. Calculate the probability that the gambler wins within 2 rounds.
- (10 points) Let p_i denote the probability that the gambler eventually succeeds before going bankruptcy, starting from position i . Determine the values of p_i for $i = 1, 2, 3$.
- (10 points) Let μ_i denote the expected number of rounds until the game ends, starting from i . Derive the recursive formula for μ_i and specify boundary conditions. (No need to solve the equations for their values.)

3. (Serving Customers) At Tiny World Coffee, customers arrive according to a Poisson process with a rate of two customers every minute. Each customer, independently of anything else, orders a cup of coffee with probability 0.4, orders a cup of non-caffeinated drinks with probability 0.4, and orders a chocolate cookie with probability 0.2. For simplicity, let's assume that each customer orders one and only one item.
- (a) (10 points) On average, how many cups of coffee are sold every minute?
 - (b) (10 points) What is the probability that exactly 4 cups of coffee have been sold before the first chocolate cookie is sold? (In this part, the selling of non-caffeinated drinks does not matter.)
 - (c) (10 points) Given the information that, by the time the first chocolate cookie is sold, exactly 4 cups of non-caffeinated drinks and 6 cups of coffee have been sold, what is the expected time for the first chocolate cookie to be sold?